

Question	Marking Guidance	Mark	Comments
04.1	Automarked question – ☒ Reduction of pyruvate (Box 3)	1 (AO1)	

Question	Marking Guidance	Mark	Comments
04.2	1. <u>Oxygen/O₂</u> taken up/used (by seeds); 2. Carbon dioxide/CO ₂ (given out) is absorbed by solution/potassium hydroxide/KOH; 3. Decrease in pressure/volume (of air inside);	3 (3 x AO2)	3. Ignore 'negative pressure' but reject reference to vacuum. 3. Accept 'air pressure higher than inside'.

Question	Marking Guidance	Mark	Comments
04.3	1. Distance (drop/liquid moves); 2. Diameter/radius/bore of tubing/lumen;	2 (2 x AO2)	Ignore time. 1. Accept description of distance, e.g. 'start and end position'. 2. Accept (cross-sectional) area of tubing/lumen.

Question	Marking Guidance	Mark	Comments
04.4	1. Remove potassium hydroxide/KOH OR Remove solution which removes carbon dioxide. OR Replace potassium hydroxide/KOH with water; 2. Record distance liquid moves (without potassium hydroxide); 3. Use difference in distance liquid moves (with potassium hydroxide and without potassium hydroxide) OR Use difference in (calculated) volumes (with potassium hydroxide and without potassium hydroxide);	3 (3 x AO3)	Answers which add/use a syringe, reject mp2 and mp3. 1. Reject if seeds removed or another organism used. 2. Reject moves to the right. 2. Accept 'liquid would not move'. 3. Accept 'if liquid does not move (volume of) carbon dioxide produced is the same as (volume of) oxygen used'.

Question	Marking Guidance	Mark	Comments
04.5	Answer in the range 3×10^{-7} to 3.33×10^{-7} ;	1 (AO2)	Accept equivalent answers in this range which are not in standard form. Accept 3.0×10^{-7} Ignore any numbers after 3.33.

